How SleepTwin works

From open data to a doctor's decision - left to right in four stages

1 OPEN DATA

real & anonymized - for calibration

CDC NHANES 2017-2020

Real anonymized US adults → realistic, correlated health records

LifeSnaps (Fitbit)

Real volunteers' sleep, heart-rate & HRV patterns

Published science

Bergman glucose model, sleep → insulin effect, gene frequencies

calibrates →

2 VIRTUAL PATIENTS

the simulator - 300 patients × 21 days

Health record

Age, diagnoses + years, labs, medicines, genetic markers

Wearable stream

Sleep stages, HRV, heart rate, steps, oxygen dips

Body model

Sleep debt → insulin sensitivity → glucose every 5 min; morning BP

daily data →

3 TWIN BRAIN

AI / machine learning

Learn the person's normal

Days 1-14 set a personal baseline

Predict (LightGBM)

Tomorrow's glucose spike & BP surge, plus a glucose alert 2 hours ahead

Explain + What-If

SHAP → reasons in plain English; tests a fix in ~15 ms

risk + why →

4 DOCTOR DASHBOARD

website - laptop, tablet or phone

FastAPI service

Secure API that serves risks, reasons and trends

5 pages

Patients • Overview • Trends • What-If • How It Works

Doctor decides

Decision support only - never a diagnosis

✓ Checked on real data

In 38 real Fitbit users, short nights pushed resting heart rate up and HRV down versus each person's own normal (the idea the twin is built on).

What-If loop: the doctor tests a fix (sleep longer, earlier bedtime, CPAP) → the twin re-runs the body model + AI and shows the new risk

All patients are SIMULATED • Model-based projection for clinical decision support. Research prototype. • Team WeBareBears · PCCOE